POPULAR SYSTEMS®

(An IATF16949:2016, ISO-9001:2015,ISO 14001:2015,OHSAS18001:2007 Certified Organisation)

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APPLICATION NO:25

INTERNSHIP TRAINING REPORT ON

CONTROL PANEL MANUFACTURING AND WIRING HARNESS

SUBMITTED BY

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B.E ELECTRICAL AND ELECTRONICS ENGINEERING



M.KUMARASAMY COLLEGE OF ENGINEERING
THALAVAPALAYAM, KARUR-639 113.

(An autonomous Institution, Affiliated to Anna University, Chennai)

DECLARATION

We hereby declare that the internship training report titled **" CONTROL PANEL MANUFACTURING AND WIRING HARNESS" in POPULAR SYSTEM PRIVATE LIMITED**, submitted to the **DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING** in **M.KUMARASAMY COLLEGE OF ENGINEERING KARUR** for awarding the degree of **BACHELOR OF ENGINEERING** is a reward of original work done by us under the guidance of **Mrs. C. SUGUNA, Mrs. M. KANAGARATHINAM.**

This report is only for reference and no part of the report will be published or copied anywhere without permission from **POPULAR SYSTEM PRIVATE LIMITED** officials.

SIGNATURE OF THE STUDENTS

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DATE:29-07-2026
PLACE: COIMBATORE

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INTRODUCTION

Background of the Company:

The primary objective of **M/S POPULAR SYSTEMS** is to manufacture high quality Assembly **Wiring Harnesses, Junction boxes, Sub-assemblies, Control panels & Wiring accessories.** The main agenda of the company since inception has been to provide its customers with world class products at competitive prices and fortify consumer trust. Since its inception, ECC has been committed to enhancing energy efficiency and sustainability by offering cutting-edge technologies, expert consultancy, and customized energy solutions. The company focuses on power conditioning, solar energy solutions, and energy audits to help clients optimize their energy consumption and reduce operational costs.

(i) Key Areas of Expertise:

Commitment to the Focused Factory (Synchronous Manufacturing) concept has allowed us to streamline manufacturing in many ways, including

- Better Workflow
- Improved Material Handling
- Reduced Inventory
- Better utilization of Floor space
- Commitment to Quality and Innovation

Our diverse manufacturing capabilities include

- Wiring Harness
- Panel assemblies
- Other Custom made contract manufactured parts.

(ii) Vision and Mission:

Our commitment to **Total Quality Management** is supported by **Strategic planning, Process analysis and Implementation**. From the moment RFQ is received, a cross-functional team is dedicated to needs from product definition, through launch and on to program assessment where we identify opportunities for continuous improvement

PURPOSE OF THE INTERNSHIP:

- **Business Development**: Understanding market opportunities and strategies to achieve leadership in the industry.
- **Marketing & Branding**: Supporting product presentations, enforcing pricing structures, and assisting in promotional plans.
- **Technical Knowledge**: Learning about **electrical, electronics, and automation wiring harness products**.
- **Product Development**: Contributing to new product launches and understanding customer and market scenarios.

Key Features of the company:

- **Industry Expertise:** They manufacture wiring harnesses, junction boxes, control panels, and wiring accessories for sectors like automotive, renewable energy, medical equipment, and home appliances.
- **Precision Engineering:** The company focuses on cutting-edge technology and synchronous manufacturing, ensuring efficient workflow, material handling, and inventory management.
- **Global Reach:** They cater to OEMs and international clients, providing customized wiring solutions.
- **Established Presence:** Founded in 2004, Popular Systems has built a strong reputation in the electrical and electronics industry.

SCOPE OF WORK:

Popular Systems operates in the **electrical and electronics industry**, specializing in **wiring harnesses, junction boxes, control panels, and wiring accessories**. The scope of work in the company includes:

- **Manufacturing & Engineering**: Designing and producing customized wiring solutions for industries like automotive, renewable energy, medical equipment, and home appliances.
- **Quality Assurance**: Ensuring high standards in product development through rigorous testing and compliance.
- **Business Development & Sales**: Expanding market reach by working with OEMs and international clients.
- **Research & Innovation**: Developing new technologies to improve efficiency and reliability in wiring systems.
- **Supply Chain & Logistics**: Managing inventory, material handling, and workflow optimization.

CONTROL PANEL ASSEMPLY OVERVIEW

Popular Systems: Leading Manufacturer & Exporter of Electrical Control Panels:

Popular Systems is a prominent manufacturer and supplier of electrical control panels, recognized for its high-quality, durable materials and environmentally friendly processing techniques. With expertise in custom-built solutions, the company serves diverse industries, including pumps, CNC lathes, and Special Purpose Machines (SPMS).

At Popular Systems, the focus is not only on manufacturing control panels but on delivering performance-driven solutions that enhance efficiency, safety, and reliability worldwide.

KEY COMPONENTS OF PANEL BOARD ASSEMBLY:

A control panel is a device that manages, monitors, and controls industrial equipment and processes. It contains various electrical components, including input devices, output devices, control logic, communication interfaces, power supplies, and the panel's enclosure.

Key Components:

Input Devices:

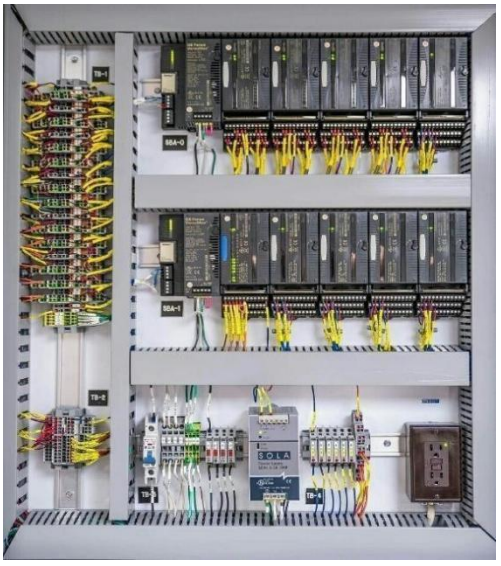
These allow operators to interact with the control panel and initiate actions, such as push buttons, toggle switches, and potentiometers.

Output Devices:

These respond to the commands from the operator, including relays, solenoids, and actuators.

Control Logic:

This is "brain" of the control panel, responsible for interpreting inputs and executing outputs. This can be achieved through relays, Programmable Logic Controllers (PLCs), or other controllers.



Programmable Logic Controller

Communication Interfaces:

These facilitate communication between the control panel and other systems or devices, including RS-232, RS-485, Ethernet, and Modbus.



Industrial Control Panel

Power Supply:

Provides the necessary power to operate the components within the control panel, including both AC and DC power supplies and sometimes battery backup systems.

Enclosure:

The physical box or cabinet that houses all the components, providing protection and organization.



Top Enclosure

Other Important Components:

Switches: Used to interrupt or change the state of electrical circuits.



Switches with Indicating LED's

Indicators: Provide visual feedback on the status of the system, such as lights or displays.

Circuit Breakers: Protect electrical circuits from overloads and short circuits by interrupting current flow.



Circuit Breaker

Relays: Electrically or magnetically operated switches that control larger loads based on a smaller control signal.



Relay Panel

Contactors: Similar to relays, but designed for larger loads.

Terminal Blocks: Provide a connection point for wires, facilitating wiring within the panel.

Human Machine Interface (HMI): Allows operators to interact with the control panel and monitor system status.



Motor Drives: Control the operation of motors, including speed, torque, and direction.

WIRING HARNESS

When discussing the wire harness manufacturers in India, many Wiring loom manufacturers exist. However Popular Systems give a wide scope of top notch Wiring Harness that are produced from the ideal quality raw materials. To be as the leading Cable loom manufacturers in India , we have established our plants in Chennai, Bangalore and Hosur. Regardless of the industry in which they are being used, their wire harnesses and cables are designed with utmost precision.Wire Harness given by us is accessible in different sizes, measurements and hues to meet the required necessities of the customers.

Popular Systems possess advanced infrastructural facilities, which are competent enough to supply bulk products in the shortest possible time period. Our wiring assemblies are very tough and can provide high wear and tear quality. We have carved ourselves as one of the recognized Wiring Harness Manufacturers and Wire Harness Suppliers, in India.

We provide a wide range of high quality automotive Wiring harness connectors that are manufactured from the top-notch quality raw materials. Our custom wiring harnesses or Cable Harnesses are widely used in industries such as automobile, elevators, escalators, textile machinery, electronics, submersible pumps, and junction boxes , providing reliable electrical connections for seamless operations.

As a reputed Wiring Harness Company in India, Popular Systems is committed to delivering top-notch cable harness solutions that enhance performance and reliability across industries. Whether you need automotive wiring harness connectors, custom wiring harnesses, or cable looms, we provide Cost-effective and efficient solutions with superior quality.

Variety of OEMs as a contract manufacturer of Wiring harness. We provide harnesses for various sectors of 2 wheeler/wheelers and 4 wheelers like

1. Dash Boards
2. Horns
3. Melody maker
4. Head lamps
5. Steering control
6. Fuel pumps
7. Tank units
8. Complete Wiring harness for Electric Bikes
9. Two wheeler wiring harness of various standard bikes
10. Battery cables



Industrial Control Panel Wiring Harness



Electronic Control Device Wiring Harnesses



Automotive Wiring Harness connectors

UPS ASSEMBLY EATON

Popular Systems has partnered with Eaton Power Quality Pvt. LTD., a globally recognized leader in power management solutions, to provide innovative UPS systems that ensure uninterrupted operations for businesses.

Eaton is a trusted name in power management, known for delivering high-performance solutions that keep businesses running efficiently. Their expertise helps industries enhance productivity, protect critical infrastructure, and stay ahead of competitors.

Key Power Solutions Offered

Popular Systems, in collaboration with Eaton, provides a range of power management products, including:

- **Uninterruptible Power Supplies (UPS)**: Ensures consistent and reliable power for critical operations, preventing downtime.
- **Power Distribution Units (PDU)**: Efficiently distributes power across multiple devices, optimizing energy usage.
- **Surge Protection Devices (SPD)**: Safeguards electrical systems from voltage spikes and power surges, ensuring longevity.

Benefits of Eaton's Technology:

- Reduce downtime by maintaining a stable power supply.
- Cut energy costs through efficient power management.
- Extend equipment lifespan with advanced protection mechanisms.
- Ensure seamless operations with reliable infrastructure support.



Technical Expertise & Engineering Capabilities

At Popular Systems specializes in **designing, manufacturing, and integrating electrical components** with a focus on **efficiency, safety, and reliability**. The company's technical experience includes:

1. Wiring Harness Engineering

- **Custom Wiring Solutions:** Engineers design and develop tailored wiring harnesses for automotive, industrial, and medical applications.
- **Material Selection & Testing:** The company ensures high-quality insulation, connectors, and terminals for durability and performance.
- **Automation & Precision Assembly:** Advanced automated processes enhance accuracy and consistency in wiring harness production.

2. **Electrical Control Panel Manufacturing**

- **Custom-Built Control Panels:** Designed for specific industrial applications, ensuring optimal performance.
- **Integration of Key Components:** Panels include contactors, circuit breakers, push buttons, switches, meters, and digital displays.
- **Safety & Compliance:** Engineers follow strict quality control measures to meet international safety standards.

3. Power Management Solutions

- **UPS Systems & Surge Protection**: Popular Systems collaborates with Eaton Power Quality Pvt. LTD. to provide advanced power management solutions.
- **Energy Efficiency & Sustainability**: The company focuses on reducing downtime, optimizing energy consumption, and extending equipment lifespan.

Advanced Manufacturing & Quality Assurance

Popular Systems operates a state-of-the-art manufacturing facility, equipped with modern machinery and automated systems. The company's technical experience extends to:

1. Precision Engineering & Automation

- **CNC Machining & Automated Assembly**: Ensures high accuracy in component production.
- **Robotic Process Automation (RPA)**: Enhances efficiency and reduces human error in manufacturing.

2. Quality Control & Testing

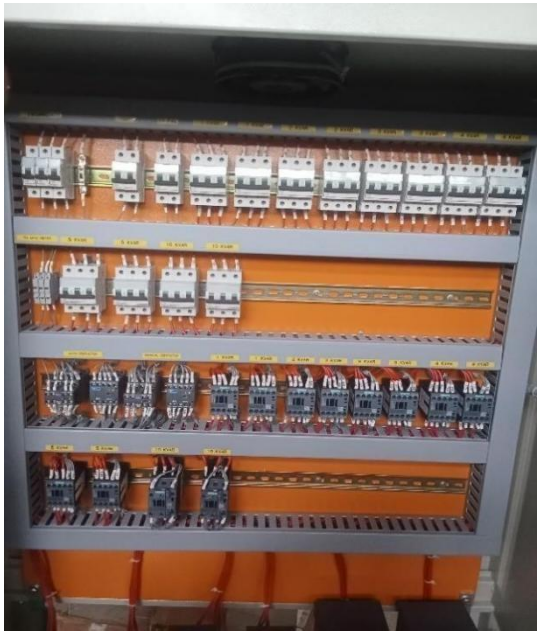
- **Rigorous Testing Procedures**: Electrical components undergo voltage, insulation, and durability tests.
- **Compliance with Industry Standards**: Products meet ISO, IEC, and other global certifications

3. Research & Development (R&D)

- **Innovation in Electrical Systems**: Engineers work on next-generation wiring solutions.
- **Prototyping & Product Development**: The company invests in new technologies to improve performance and reliability.


TRINITY
ACCUVAR






DANGER

LEARNING OBJECTIVES

Technical Learning Objectives

- **Understanding Wiring Harness Engineering**: Learn about design, material selection, and automated assembly for wiring harnesses used in automotive, industrial, and medical applications.
- **Electrical Control Panel Development**: Gain expertise in custom-built control panels, integrating contactors, circuit breakers, push buttons, and digital displays.
- **Power Management Solutions**: Explore UPS systems, power distribution units (PDUs), and surge protection devices (SPDs) in collaboration with Eaton Power Quality Pvt. LTD..
- **Quality Assurance & Testing**: Learn rigorous testing procedures for electrical components, ensuring compliance with industry standards.

Business & Industry Learning Objectives

- **Market Analysis & Business Development:** Understand customer needs, pricing strategies, and branding in the electrical solutions market.
- **Supply Chain & Logistics:** Learn about inventory management, material handling, and workflow optimization.
- **Project Management & Innovation:** Develop skills in product development, research, and implementation of new technologies.

CONCLUSION

The internship at Popular Systems has been a highly enriching experience, offering both technical and professional growth in the electrical and electronics industry. Interns gain hands-on exposure to wiring harness engineering, electrical control panels, and advanced power management solutions. The company fosters a structured learning environment where interns develop expertise in manufacturing, quality assurance, market strategy, and supply chain optimization.

During the internship, working with high-quality materials, modern engineering techniques, and automated processes enhances understanding of industry standards and best practices. Interns actively participate in product development, testing procedures, and real-world problem-solving, contributing to the company's innovative approach.

Beyond technical skills, the experience builds communication, teamwork, and project management abilities, equipping interns with practical insights into the business side of the industry. Exposure to customer-oriented solutions and market dynamics prepares interns for future career challenges in the engineering and manufacturing sectors.

In conclusion, the internship at Popular Systems provides a strong foundation for professional advancement. It prepares interns with the technical proficiency, industry knowledge, and strategic problem-solving skills necessary for success in the electrical and automation field. The experience is invaluable, bridging the gap between academic learning and practical application, making it a stepping stone to a promising career.